

Matter around us

1. Anything that has mass and occupies space is called

- A) Energy
- B) Matter
- C) Force
- D) Heat

Answer: B

2. Matter is made up of

- A) Atoms and molecules
- B) Cells
- C) Light
- D) Energy

Answer: A

3. The particles of matter are

- A) Stationary
- B) Very small
- C) Very large
- D) Invisible only

Answer: B

4. Matter exists in

- A) One state
- B) Two states
- C) Three states
- D) Four states

Answer: C

5. The three common states of matter are

- A) Solid, liquid, gas
- B) Solid, plasma, gas
- C) Liquid, plasma, gas
- D) Solid, liquid, plasma

Answer: A

6. In solids particles are

- A) Very far apart
- B) Very close together
- C) Moving freely
- D) Not moving

Answer: B

7. Solids have

- A) Definite shape and volume
- B) No shape
- C) No volume
- D) Variable shape

Answer: A

8. Liquids have

- A) Definite shape and volume
- B) No shape but definite volume
- C) No volume
- D) No shape and no volume

Answer: B

9. Gases have

- A) Definite shape
- B) Definite volume
- C) Neither definite shape nor volume
- D) Fixed shape only

Answer: C

10. The particles of matter are always

- A) At rest
- B) In motion
- C) Fixed
- D) Heavy

Answer: B

11. The movement of particles in matter is called

- A) Diffusion

- B) Motion
- C) Vibration
- D) Brownian motion

Answer: A

12. Diffusion is faster in

- A) Solids
- B) Liquids
- C) Gases
- D) Metals

Answer: C

13. Diffusion occurs because particles

- A) Are stationary
- B) Move randomly
- C) Are large
- D) Are fixed

Answer: B

14. The force of attraction between particles is strongest in

- A) Solids
- B) Liquids
- C) Gases
- D) Plasma

Answer: A

15. The force of attraction between particles is weakest in

- A) Solids
- B) Liquids
- C) Gases
- D) Metals

Answer: C

16. Compressibility is highest in

- A) Solids
- B) Liquids
- C) Gases

D) Metals

Answer: C

17. Compressibility is lowest in

A) Solids

B) Liquids

C) Gases

D) Air

Answer: A

18. The change of state from solid to liquid is called

A) Freezing

B) Melting

C) Condensation

D) Sublimation

Answer: B

19. The temperature at which solid changes to liquid is

A) Boiling point

B) Melting point

C) Freezing point

D) Condensation point

Answer: B

20. The change of state from liquid to gas is called

A) Boiling

B) Freezing

C) Melting

D) Condensation

Answer: A

21. The temperature at which liquid changes to gas is

A) Melting point

B) Freezing point

C) Boiling point

D) Sublimation point

Answer: C

22. Conversion of gas into liquid is

- A) Freezing
- B) Condensation
- C) Melting
- D) Sublimation

Answer: B

• Conversion of liquid into solid is

- A) Freezing
- B) Boiling
- C) Melting
- D) Evaporation

Answer: A

• Direct change from solid to gas is called

- A) Condensation
- B) Sublimation
- C) Freezing
- D) Melting

Answer: B

• Example of sublimation substance is

- A) Ice
- B) Camphor
- C) Water
- D) Salt

Answer: B

• Evaporation occurs at

- A) Boiling point only
- B) Any temperature
- C) Freezing point
- D) Melting point

Answer: B

• Evaporation causes

- A) Cooling
- B) Heating
- C) Pressure increase
- D) Density increase

Answer: A

• Rate of evaporation increases with

- A) Temperature
- B) Surface area
- C) Wind speed
- D) All of these

Answer: D

- Humidity refers to
 - A) Amount of water vapour in air
 - B) Amount of oxygen
 - C) Amount of nitrogen
 - D) Air pressure

Answer: A

- High humidity
 - A) Slows evaporation
 - B) Increases evaporation
 - C) Stops evaporation
 - D) Has no effect

Answer: A

- During evaporation particles
 - A) Lose energy
 - B) Gain energy
 - C) Stop moving
 - D) Become solid

Answer: B

- The smell of perfume spreads due to
 - A) Conduction
 - B) Diffusion
 - C) Radiation
 - D) Condensation

Answer: B

- Liquids flow because
 - A) Weak intermolecular forces
 - B) Strong forces
 - C) No particles
 - D) Large particles

Answer: A

- Gas fills the container because particles
 - A) Are fixed
 - B) Move freely
 - C) Are heavy
 - D) Are cold

Answer: B

- Increasing temperature increases
 - A) Particle motion
 - B) Particle size
 - C) Particle mass
 - D) Particle colour

Answer: A

- Plasma is a state of matter consisting of
 - A) Neutral particles

- B) Ionized gas
- C) Solid ions
- D) Liquid ions

Answer: B

- Plasma is found in

- A) Sun
- B) Moon
- C) Earth core
- D) Ocean

Answer: A

- Bose–Einstein condensate forms at

- A) High temperature
- B) Very low temperature
- C) Room temperature
- D) Boiling point

Answer: B

- BEC was predicted by

- A) Newton
- B) Einstein and Bose
- C) Dalton
- D) Rutherford

Answer: B

- Solid particles vibrate about

- A) Fixed position
- B) Random position
- C) Free space
- D) Container walls

Answer: A

- Liquids take the shape of

- A) Container
- B) Their own shape
- C) Fixed shape
- D) No shape

Answer: A

- Gas pressure is due to

- A) Particle collisions
- B) Gravity
- C) Heat only
- D) Colour

Answer: A

- Matter expands when

- A) Heated
- B) Cooled
- C) Pressed

D) Frozen

Answer: A

- Matter contracts when

A) Heated

B) Cooled

C) Pressed

D) Melted

Answer: B

- Solid to gas change without liquid is

A) Sublimation

B) Freezing

C) Condensation

D) Melting

Answer: A

- Dry ice is

A) Solid oxygen

B) Solid carbon dioxide

C) Solid nitrogen

D) Solid hydrogen

Answer: B

- Dry ice changes directly to

A) Liquid

B) Gas

C) Plasma

D) Solid

Answer: B

- Diffusion in liquids is

A) Faster than gases

B) Slower than gases

C) Same as gases

D) Impossible

Answer: B

- Particles of matter attract each other due to

A) Intermolecular forces

B) Gravity only

C) Pressure

D) Energy

Answer: A

- Matter occupies

A) Space

B) Light

C) Energy

D) Force

Answer: A

- Solid particles are arranged

A) Randomly
B) Closely packed
C) Very far
D) Free moving

Answer: B

- Liquids have

A) Fixed shape
B) No volume
C) Fixed volume
D) Fixed shape and volume

Answer: C

- Gas particles move

A) Slowly
B) Randomly and freely
C) Fixed
D) Not moving

Answer: B

- Interparticle space is maximum in

A) Solids
B) Liquids
C) Gases
D) Metals

Answer: C

- Interparticle space is minimum in

A) Solids
B) Liquids
C) Gases
D) Air

Answer: A

- Change of state depends on

A) Temperature
B) Pressure
C) Both
D) Colour

Answer: C

- Increasing pressure can convert

A) Gas to liquid
B) Solid to gas
C) Liquid to solid
D) Plasma to gas

Answer: A

- Decreasing temperature causes

A) Condensation

- B) Evaporation
- C) Melting
- D) Sublimation

Answer: A

- Evaporation takes place at

- A) Surface
- B) Bottom
- C) Middle
- D) Inside only

Answer: A

- Cooling effect of evaporation is used in

- A) Desert coolers
- B) Heaters
- C) Lamps
- D) Fans

Answer: A

- Cotton clothes are preferred in summer because

- A) Absorb sweat
- B) Increase heat
- C) Reflect light
- D) Produce heat

Answer: A

- Diffusion rate increases with

- A) Temperature
- B) Pressure
- C) Density
- D) Colour

Answer: A

- Gas molecules have

- A) High kinetic energy
- B) Low kinetic energy
- C) No energy
- D) Same energy as solids

Answer: A

- Solid particles have

- A) Lowest kinetic energy
- B) Highest kinetic energy
- C) Medium energy
- D) No energy

Answer: A

- Liquid particles have

- A) Medium kinetic energy
- B) Zero energy
- C) Highest energy

D) Lowest energy

Answer: A

- The state with fixed shape is

A) Solid

B) Liquid

C) Gas

D) Plasma

Answer: A

- The state with highest compressibility

A) Solid

B) Liquid

C) Gas

D) Metal

Answer: C

- Evaporation rate decreases when

A) Humidity increases

B) Temperature increases

C) Surface area increases

D) Wind increases

Answer: A

- Particle motion increases with

A) Temperature

B) Pressure decrease

C) Heat

D) All of these

Answer: D

- Liquids can be compressed

A) Easily

B) Slightly

C) Completely

D) Fully

Answer: B

- Solids cannot be compressed because

A) Particles close together

B) No particles

C) High energy

D) No attraction

Answer: A

- Gas molecules are

A) Far apart

B) Close

C) Fixed

D) Packed

Answer: A

- Diffusion proves that particles

A) Are moving
B) Are stationary
C) Are large
D) Are visible

Answer: A

- Sublimation occurs in

A) Iodine
B) Iron
C) Copper
D) Gold

Answer: A

- Example of solid is

A) Water
B) Air
C) Ice
D) Oxygen

Answer: C

- Example of liquid is

A) Water
B) Ice
C) Oxygen
D) Carbon dioxide

Answer: A

- Example of gas is

A) Oxygen
B) Ice
C) Salt
D) Sugar

Answer: A

- Particles of matter are

A) Very small
B) Visible
C) Large
D) Heavy

Answer: A

- Matter can change state due to

A) Temperature change
B) Pressure change
C) Both
D) None

Answer: C

- Energy required for state change is

A) Latent heat

- B) Kinetic energy
- C) Potential energy
- D) Mechanical energy

Answer: A

- Boiling occurs
- A) Throughout liquid
- B) Surface only
- C) Bottom only
- D) Middle only

Answer: A

- Evaporation occurs
- A) Surface only
- B) Entire liquid
- C) Bottom
- D) Middle

Answer: A

- Evaporation is faster when wind
- A) Increases
- B) Decreases
- C) Stops
- D) Remains same

Answer: A

- Pressure can convert gas into
- A) Liquid
- B) Solid
- C) Plasma
- D) Vacuum

Answer: A

- Particles in gas collide with
- A) Each other and container
- B) Only container
- C) Only particles
- D) Nothing

Answer: A

- Matter expands due to
- A) Heat
- B) Cold
- C) Pressure
- D) Light

Answer: A

- Cooling causes particles to
- A) Move slower
- B) Move faster
- C) Stop

D) Increase size

Answer: A

- Matter in plasma state contains

A) Ions and electrons

B) Atoms only

C) Molecules only

D) Solids

Answer: A

- BEC is

A) Fifth state of matter

B) Third state

C) Second state

D) First state

Answer: A

- BEC occurs near

A) Absolute zero

B) Boiling point

C) Melting point

D) Room temperature

Answer: A

- Absolute zero temperature is

A) -273°C

B) 0°C

C) 100°C

D) 273°C

Answer: A

- Increasing heat causes

A) Expansion

B) Contraction

C) Freezing

D) Condensation

Answer: A

- Matter is made of

A) Tiny particles

B) Large particles

C) Only molecules

D) Only atoms

Answer: A

- Diffusion happens faster in

A) Hot gases

B) Cold solids

C) Cold liquids

D) Metals

Answer: A

- Energy of particles determines

A) State of matter
B) Colour
C) Taste
D) Smell

Answer: A

- Matter can exist as

A) Solid
B) Liquid
C) Gas
D) All of these

Answer: D

- Particles attract each other with

A) Intermolecular force
B) Gravity only
C) Pressure
D) Energy

Answer: A

- Gas particles move in

A) Random motion
B) Circular motion
C) Fixed motion
D) Straight line only

Answer: A

- When particles move faster temperature

A) Increases
B) Decreases
C) Stops
D) Remains same

Answer: A

- Matter around us exists mainly in

A) Three states
B) Two states
C) Four states
D) Five states

Answer: A